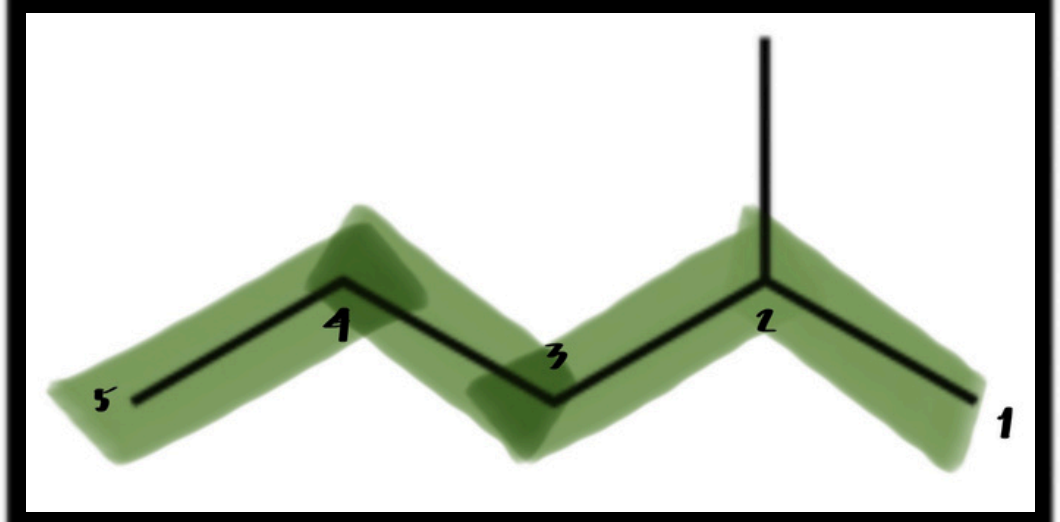
WORKED EXAMPLE A – STRUCTURE → NAME

A01+ FOUNDATION — MOLECULAR FORMULA (counted): C_6H_{14}

1. NUMBERED PARENT



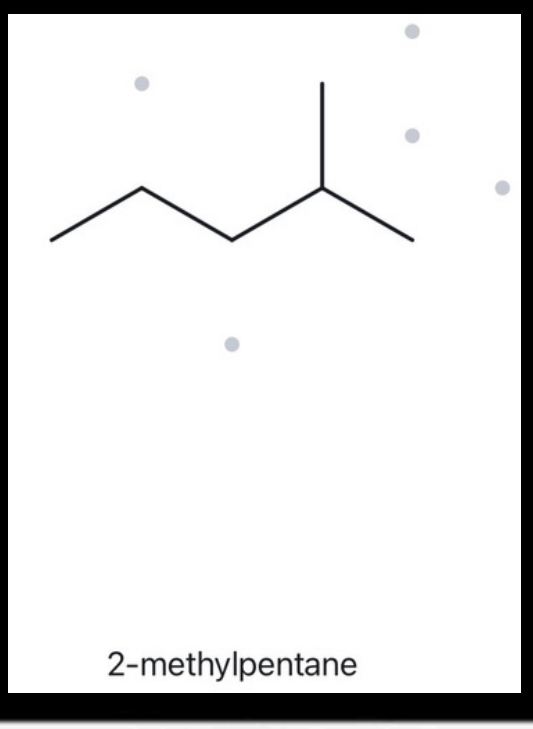
2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane. The longest continuous chain contains five carbon atoms, with one methyl branch.
Numbering: Left → right gives 4; right → left gives 2. Choose numbering from the right to obtain the lowest possible locant.
Substituents: One methyl group on C2 → 2-methylpentane.

3. MY NAME (WRITTEN FIRST)

2–methylpentane

4. THE ALKANE CHECK



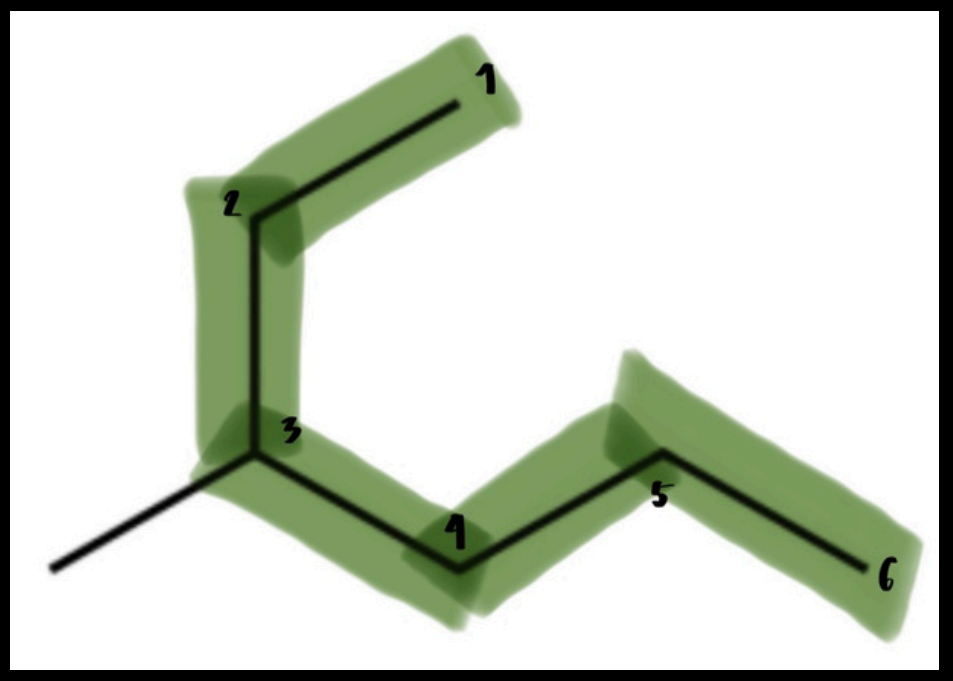
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A02 + FOUNDATION — MOLECULAR FORMULA (counted): C_7H_{16}

1. NUMBERED PARENT



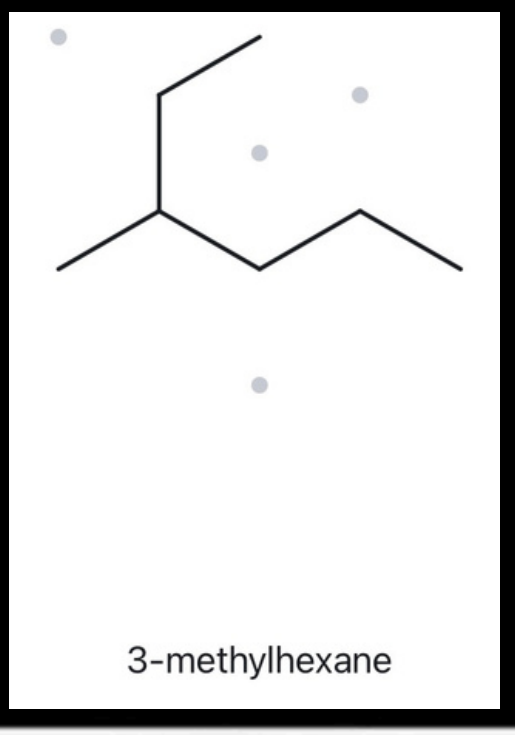
2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane. The highlighted continuous chain contains six carbon atoms, with one methyl branch attached to the parent chain.
Numbering: Top-right → bottom-left → right gives the methyl substituent at C3. Numbering from the opposite end gives C4. Choose numbering from the top-right to obtain the lowest locant (3 < 4).
Substituents: One methyl group on C3 → 3-methylhexane.

3. MY NAME (WRITTEN FIRST)

3-methylhexane

4. THE ALKANE CHECK



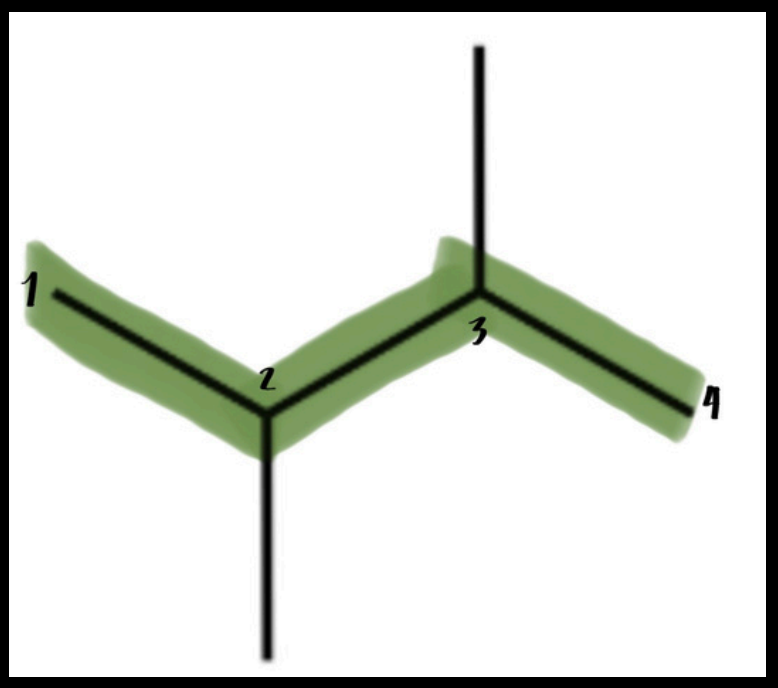
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

AO3 + FOUNDATION — MOLECULAR FORMULA (counted): C_6H_{14}

1. NUMBERED PARENT



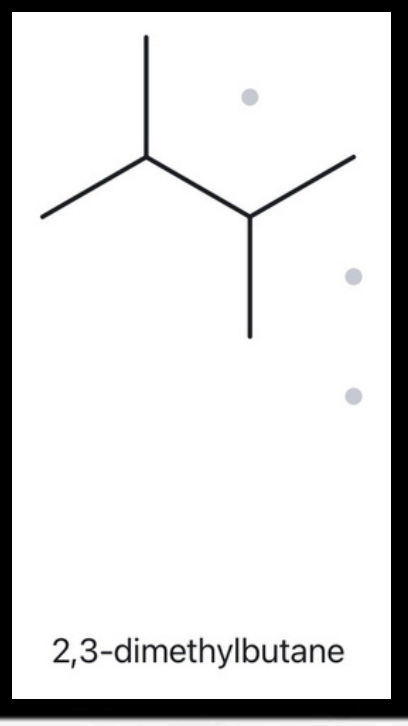
2. NAMING TRAIL

Parent: Longest chain = 4 C → butane. The highlighted continuous chain contains four carbon atoms, with two methyl branches attached to the parent chain.
Numbering: Left → right gives 2,3; right → left also gives 2,3. Both directions give the same lowest set of locants.
Substituents: Two methyl groups on C2 and C3 → 2,3-dimethylbutane.

3. MY NAME (WRITTEN FIRST)

2,3–dimethylbutane

4. THE ALKANE CHECK



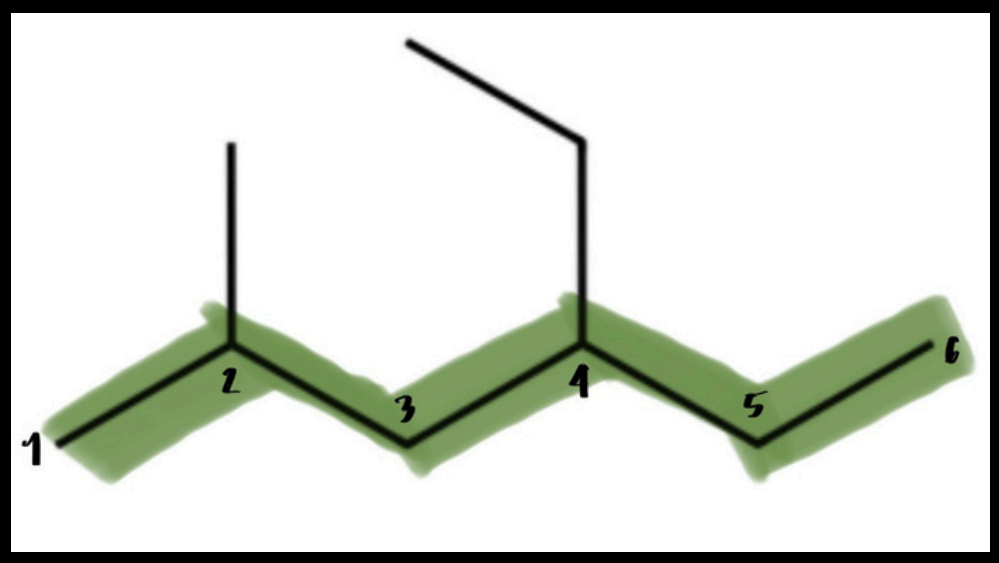
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

AO4 + DEVELOPING — MOLECULAR FORMULA (counted): C_9H_{20}

1. NUMBERED PARENT



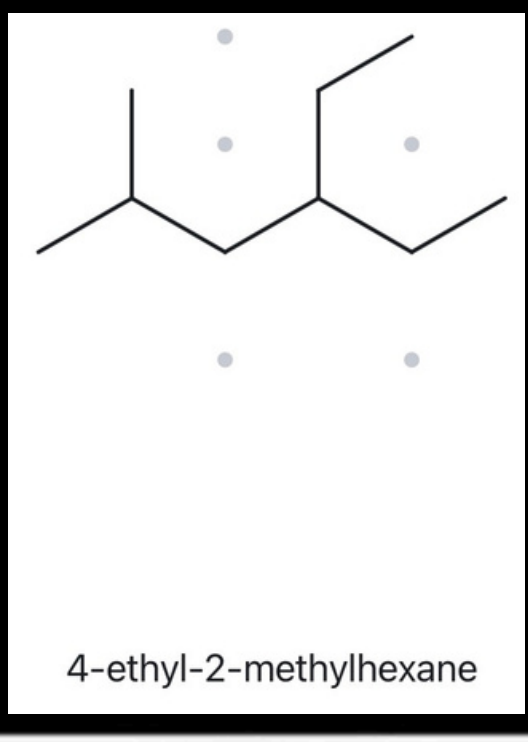
2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane. The highlighted continuous chain contains six carbon atoms.
Numbering: Left → right gives substituent positions 2,4. Right → left gives 3,5. Choose numbering from the left because 2,4 is the lower set of locants.
Substituents:
Methyl group on C2
Ethyl group on C4
Alphabetical order: Ethyl comes before methyl.
IUPAC Name: 4-ethyl-2-methylhexane

3. MY NAME (WRITTEN FIRST)

4–ethyl–2–methylhexane

4. THE ALKANE CHECK



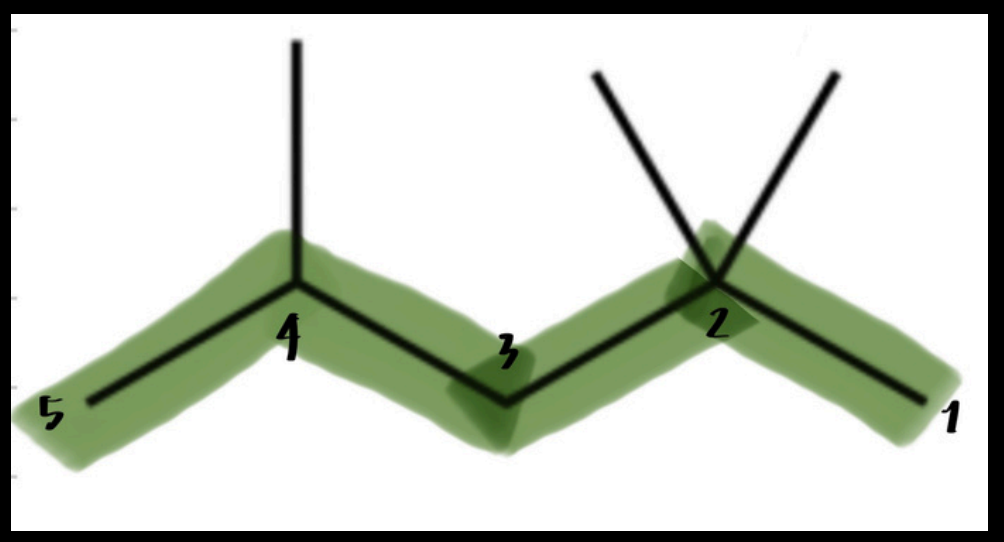
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

AOS – DEVELOPING — MOLECULAR FORMULA (counted): C_8H_{18}

1. NUMBERED PARENT



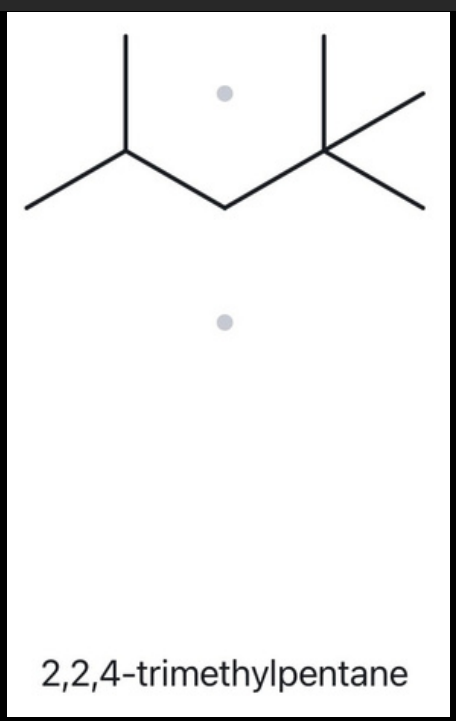
2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane. The highlighted continuous chain contains five carbon atoms, with three methyl branches attached to the parent chain.
Numbering: Right → left gives 2,2,4. Left → right gives 2,4,4. Choose numbering from the right because 2,2,4 is the lower set of locants.
Substituents:
Two methyl groups on C2
One methyl group on C4
Three methyl groups → trimethyl.
IUPAC Name: 2,2,4-trimethylpentane

3. MY NAME (WRITTEN FIRST)

2,2,4–trimethylpentane

4. THE ALKANE CHECK



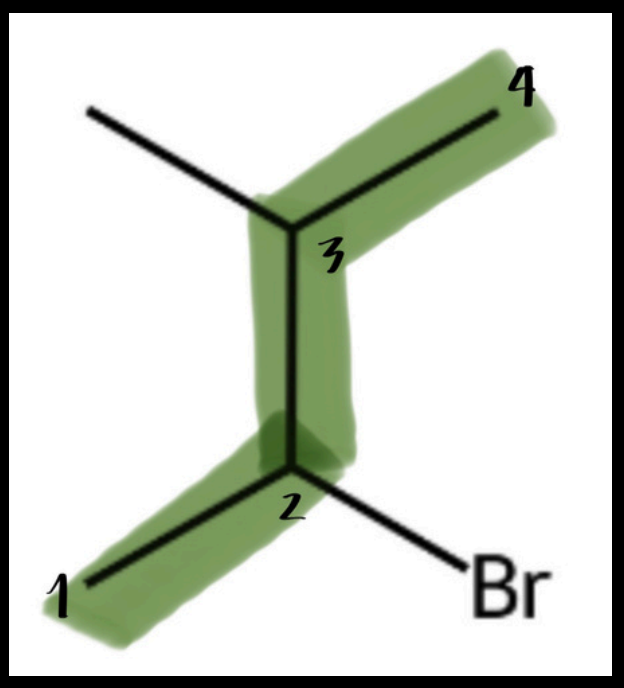
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A06 + DEVELOPING — MOLECULAR FORMULA (counted): $C_5H_{11}Br$

1. NUMBERED PARENT



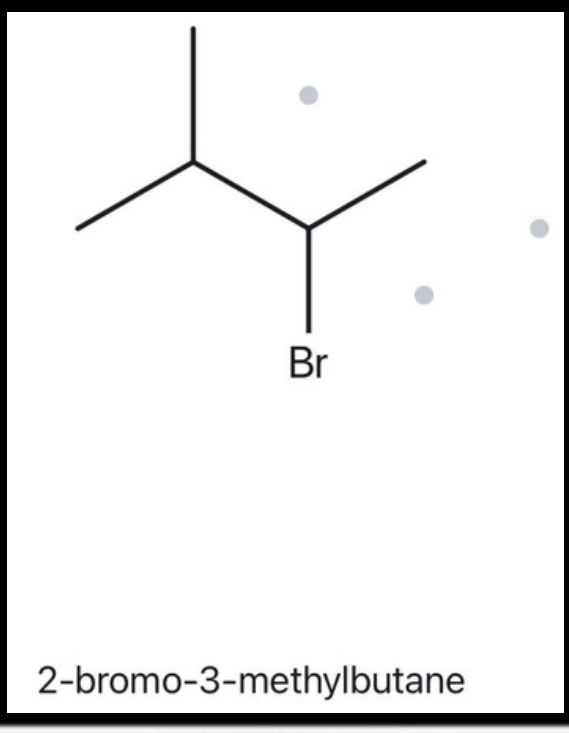
2. NAMING TRAIL

Parent: Longest chain = 4 C → butane. The highlighted continuous chain contains four carbon atoms, with one methyl branch and one bromo substituent.
Numbering:
Bottom-left → top-right gives bromo at C2 and methyl at C3.
Top-right → bottom-left gives methyl at C2 and bromo at C3.
Both give the same locant set (2,3). Apply the alphabetical tie-breaker: bromo comes before methyl, so bromo receives the lower number.
Substituents:
Bromo (Br) on C2
Methyl (CH_3) on C3
IUPAC Name: 2-bromo-3-methylbutane

3. MY NAME (WRITTEN FIRST)

2–bromo–3–methylbutane

4. THE ALKANE CHECK



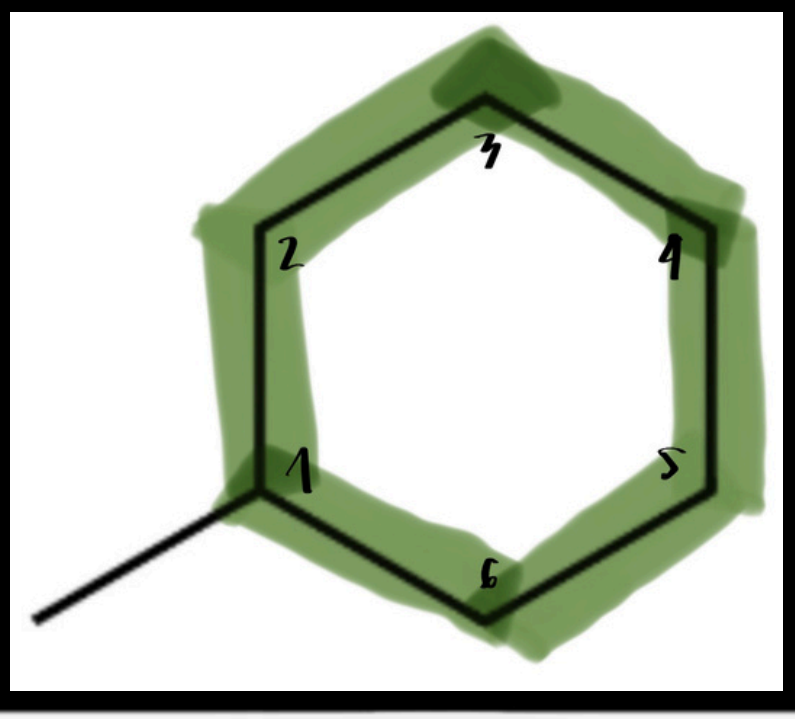
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A07 + DEVELOPING — MOLECULAR FORMULA (counted): C_7H_{14}

1. NUMBERED PARENT



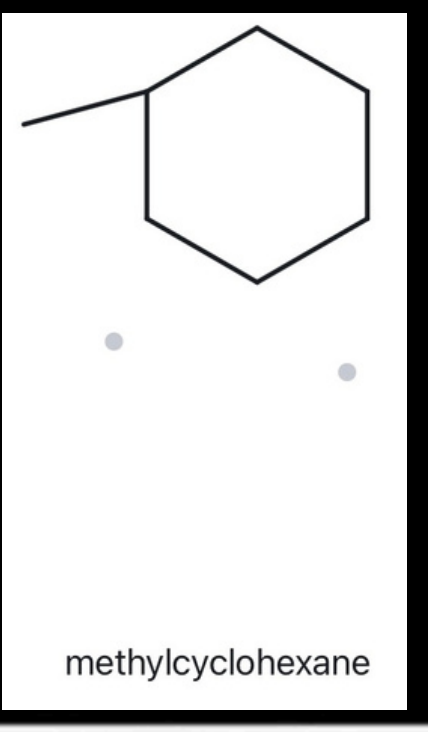
2. NAMING TRAIL

Parent: Longest continuous ring = 6 C → cyclohexane. The structure contains a six-carbon ring with one methyl substituent.
Numbering: Start numbering at the carbon attached to the methyl group (C1). Since there is only one substituent, numbering can proceed in either direction.
Substituents:
- One methyl group (CH_3) on C1.
IUPAC Name: methylcyclohexane

3. MY NAME (WRITTEN FIRST)

methylcyclohexane

4. THE ALKANE CHECK



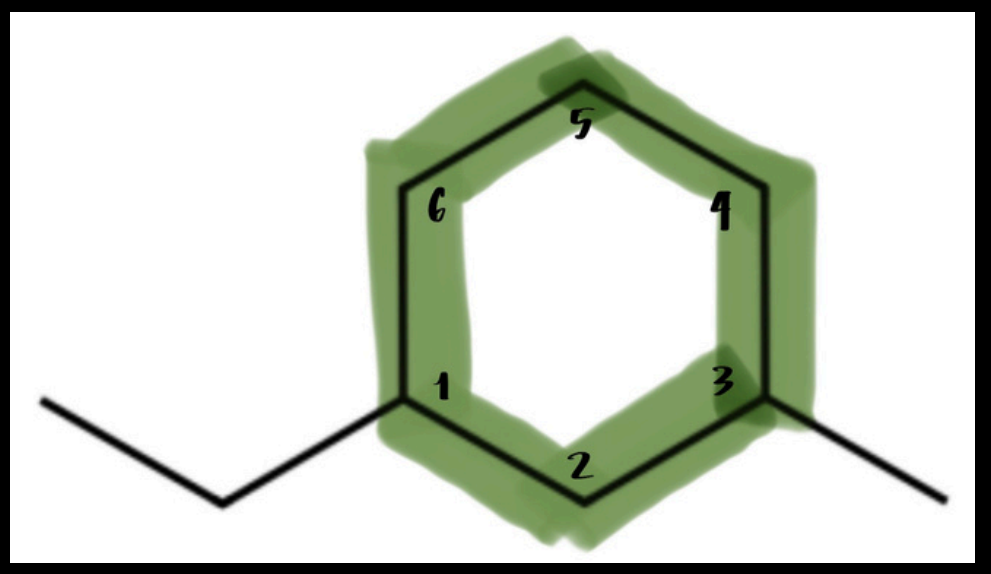
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A08 + CHALLENG — MOLECULAR FORMULA (counted): C_9H_{18}

1. NUMBERED PARENT



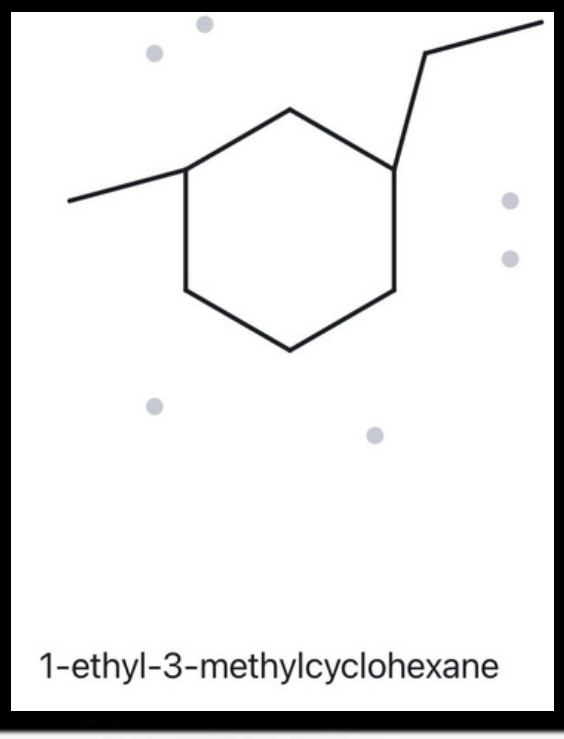
2. NAMING TRAIL

Parent: Longest ring = 6 C → cyclohexane. The structure contains a six-carbon ring with two substituents: one ethyl group and one methyl group.
Numbering:
Starting at the carbon attached to ethyl and numbering toward methyl gives positions 1,3.
Starting at the carbon attached to methyl and numbering toward ethyl also gives positions 1,3.
Since both directions give the same locant set, apply the alphabetical tie-breaker. Ethyl comes before methyl, so ethyl receives C1.
Substituents:
Ethyl group ($-CH_2CH_3$) on C1
Methyl group ($-CH_3$) on C3
IUPAC Name: 1-ethyl-3-methylcyclohexane

3. MY NAME (WRITTEN FIRST)

1–ethyl–3–methylcyclohexane

4. THE ALKANE CHECK



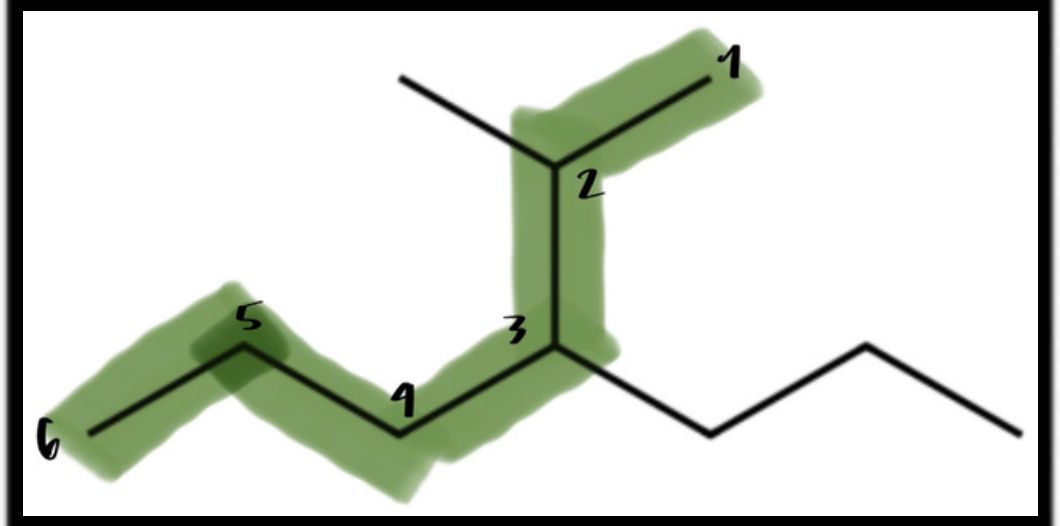
VERIFICATION STRIP

✅ longest chain ring ✅ lowest locants ✅ alphabetical order
✅ puctuation ✅ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A09 + CHALLENGE — MOLECULAR FORMULA (counted): $C_{10}H_{22}$

1. NUMBERED PARENT



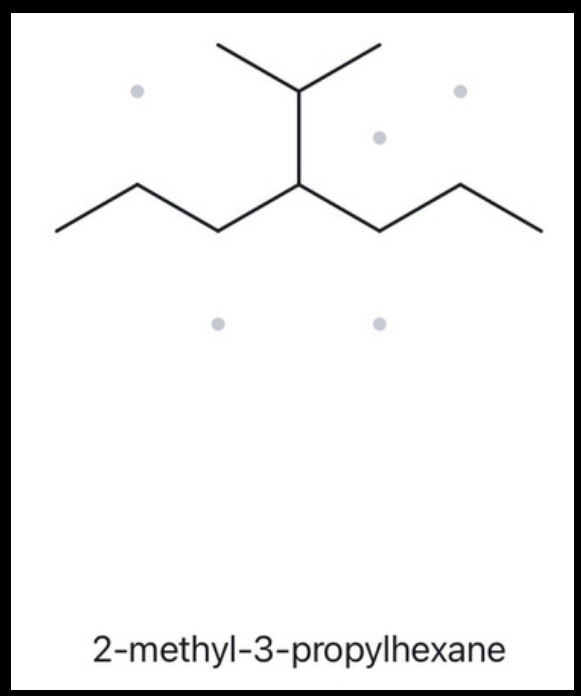
2. NAMING TRAIL

Parent: Selected chain = 6 C → hexane. The highlighted parent chain contains six carbon atoms, with two alkyl substituents attached.
Numbering:
Start from the upper-right endpoint and proceed toward the left.
This gives substituent positions 2 and 3.
Numbering from the opposite direction gives 4 and 5.
Choose 2,3 because it gives the lowest set of locants.
Substituents:
Methyl group ($-CH_3$) on C2.
Propyl group ($-CH_2CH_2CH_3$) on C3.
IUPAC Name: 2-methyl-3-propylhexane

3. MY NAME (WRITTEN FIRST)

2–methyl–3–propylhexane

4. THE ALKANE CHECK



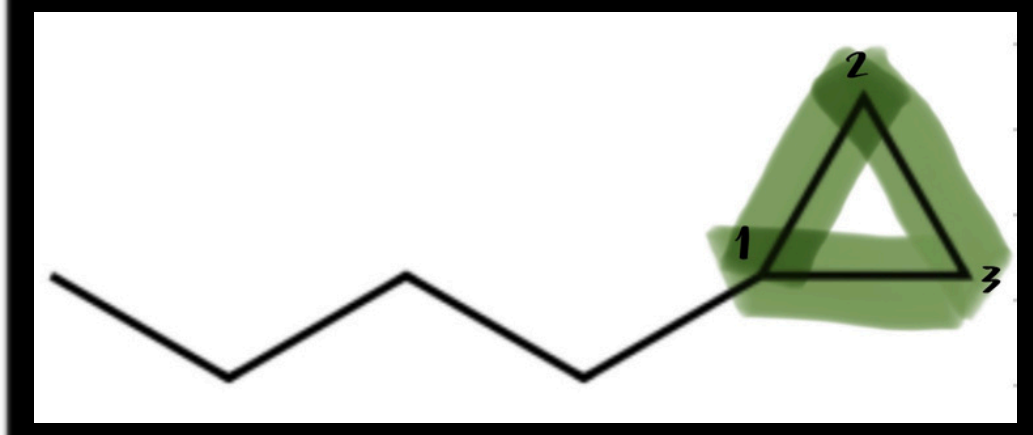
VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

WORKED EXAMPLE A – STRUCTURE → NAME

A10 + CHALLENGE — MOLECULAR FORMULA (counted): C_7H_{14}

1. NUMBERED PARENT



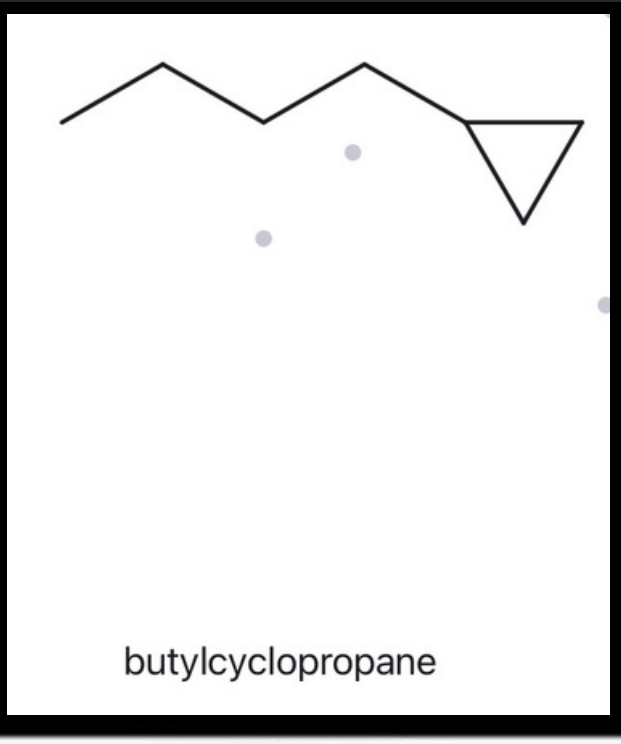
2. NAMING TRAIL

Parent: Cyclopropane ring = 3 C → cyclopropane. The three-carbon ring is selected as the parent, while the attached four-carbon straight chain is treated as a butyl substituent.
Numbering: Number the cyclopropane ring starting at the carbon attached to the butyl group (C1). Since there is only one substituent, no direction preference is necessary.
Substituents:
- One butyl group ($–CH_2–CH_2–CH_2–CH_3$) attached to C1.
IUPAC Name: butylcyclopropane

3. MY NAME (WRITTEN FIRST)

butylcyclopropane

4. THE ALKANE CHECK



VERIFICATION STRIP

☑ longest chain ring ☑ lowest locants ☑ alphabetical order
☑ puctuation ☑ formula counted

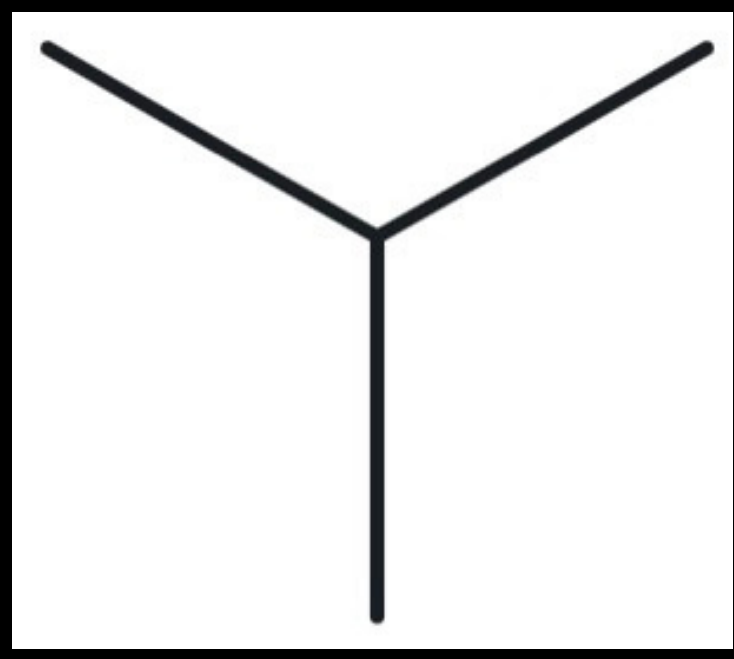
WORKED EXAMPLE B – NAME → STRUCTURE

B01 + FOUNDATION — GIVEN NAME: 2-methylpropane

1. NAME DECODED

Parent: propane → a 3-carbon chain.
Substituents: methyl on C2.
Expected formula: 3 + 1 = 4 C; no ring
→ C_nH_{2n+2} → C_4H_{10}

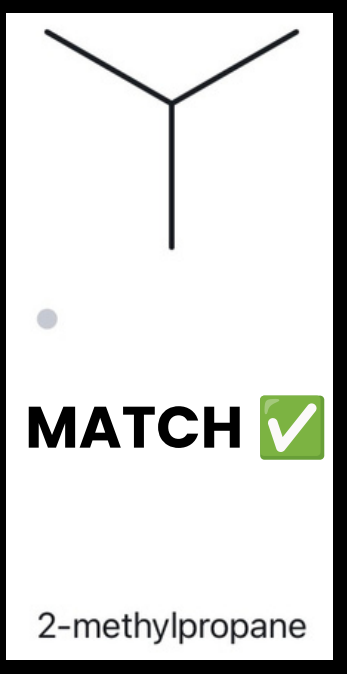
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_4H_{10}
Using the alkane formula: C_nH_{2n+2}
$C_4H_2(_4)+_2 = C_4H_{10}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

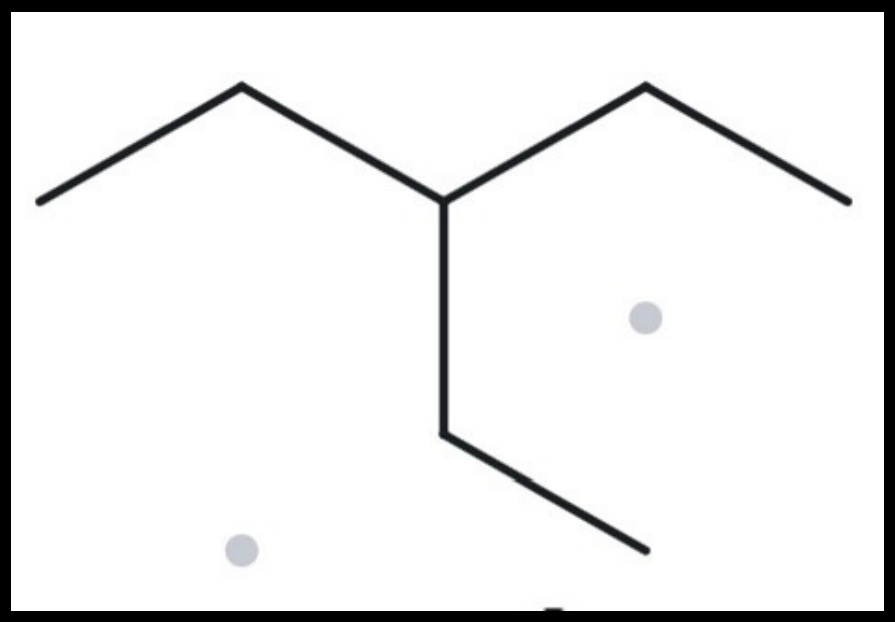
WORKED EXAMPLE B – NAME → STRUCTURE

B02 + FOUNDATION — GIVEN NAME: 3-ethylpentane

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected formula: 5 + 2 = 7 C; no ring
→ C_nH_{2n+2} → C_7H_{16}

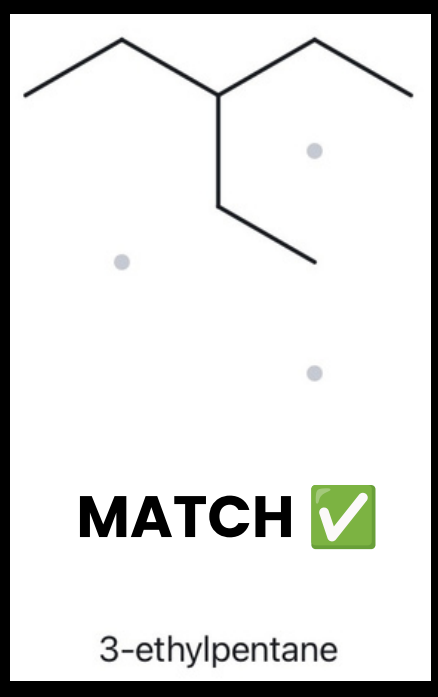
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_7H_{16}
Using the alkane formula: C_nH_{2n+2}
$C_7H_2(_7)+_2 = C_7H_{16}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

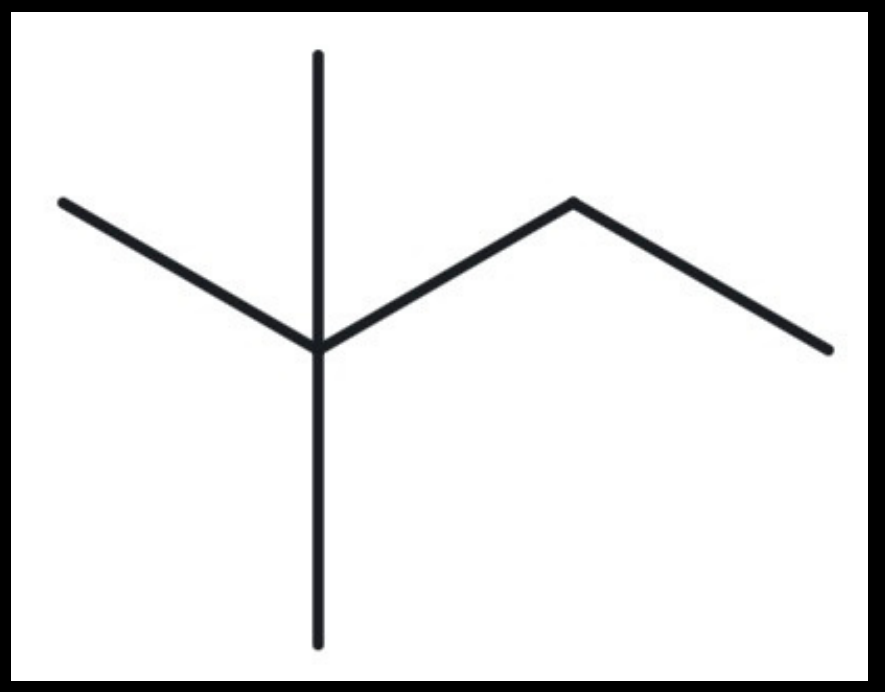
WORKED EXAMPLE B – NAME → STRUCTURE

B03 + FOUNDATION — GIVEN NAME: 2,2-dimethylbutane

1. NAME DECODED

Parent: butane → a 4-carbon chain.
Substituents: two methyl groups on C2.
Expected formula: 4 + 2 = 6 C; no ring → C_nH_{2n+2} → C_6H_{14}

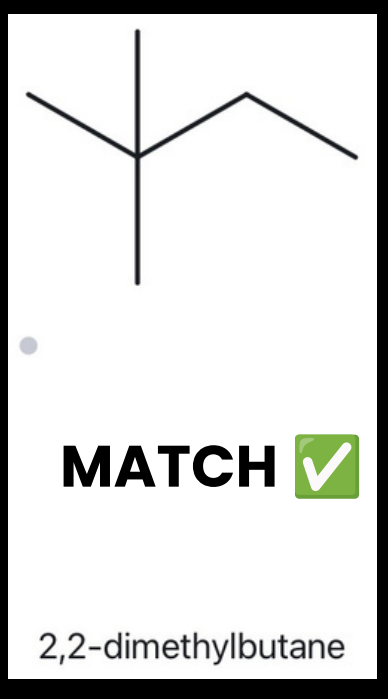
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}
Using the alkane formula: C_nH_{2n+2}
$C_6H_2(_6)+_2 = C_6H_{14}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

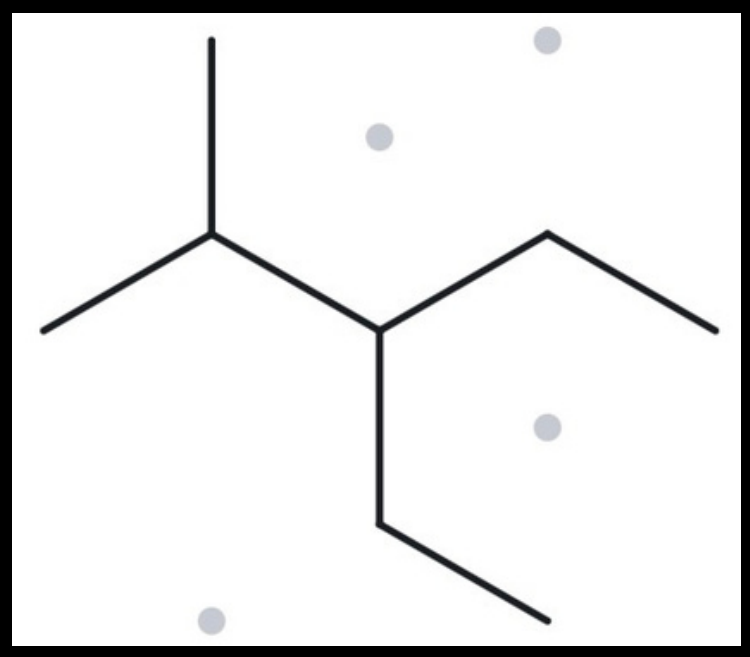
WORKED EXAMPLE B – NAME → STRUCTURE

B04 + DEVELOPING — GIVEN NAME: 3-ethyl-2-methylpentane

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: methyl on C2; ethyl on C3.
Expected formula: 5 + 1 + 2 = 8 C; no ring → C_nH_{2n+2} → C_8H_{18}

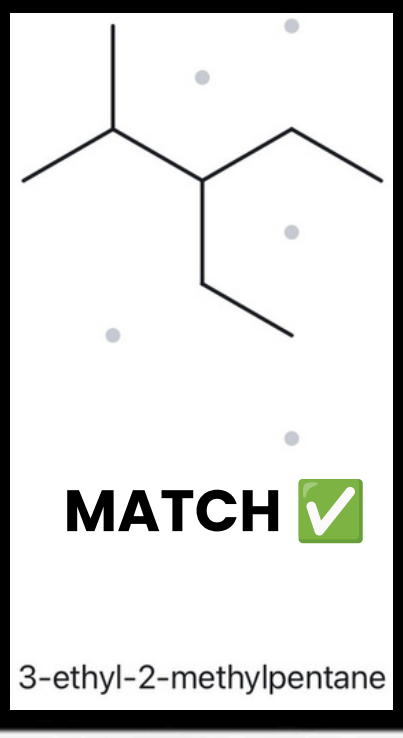
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_8H_{18}
Using the alkane formula: C_nH_{2n+2}
$C_8H_2(_8)+_2 = C_8H_{18}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

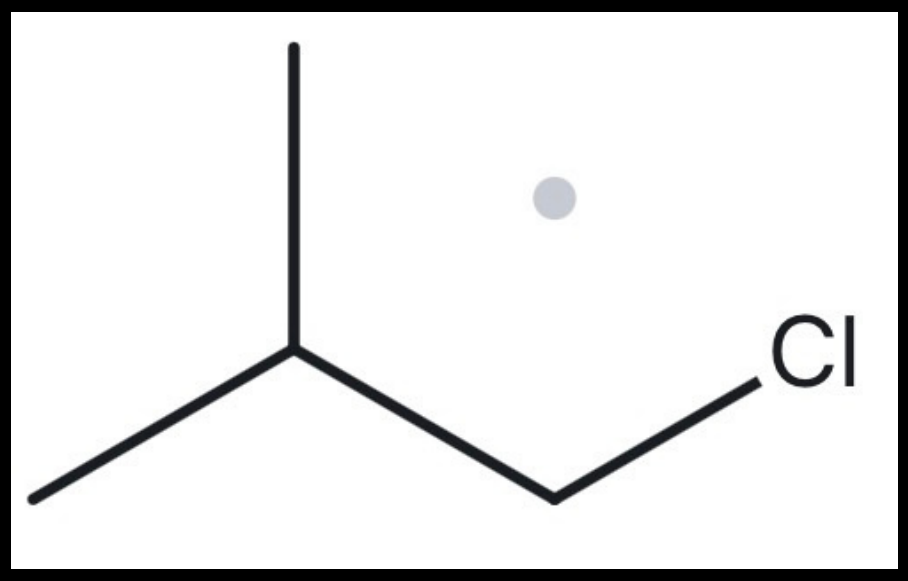
WORKED EXAMPLE B – NAME → STRUCTURE

BOS + DEVELOPING — GIVEN NAME: 1-chloro-2-methylpropane

1. NAME DECODED

Parent: propane → a 3-carbon chain.
Substituents: chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C; one chlorine replaces one hydrogen → C_4H_9Cl

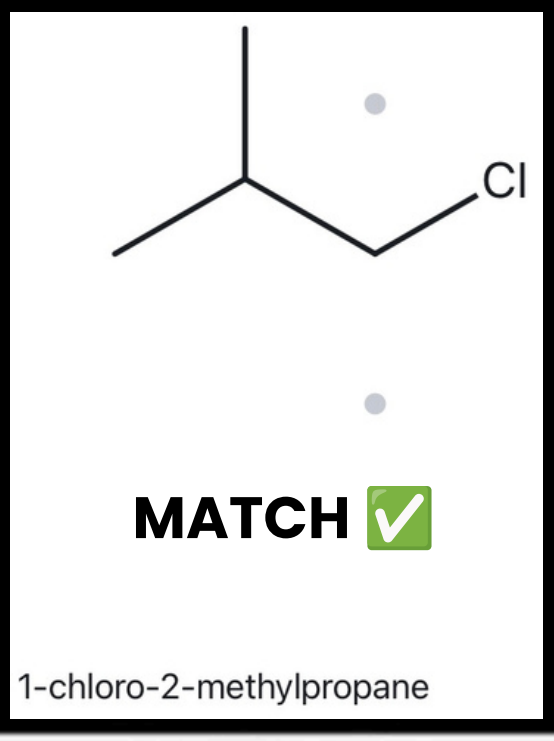
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_4H_9Cl
Using the alkane formula: C_nH_{2n+2}
$C_4H_{10} - H + Cl = C_4H_9Cl$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

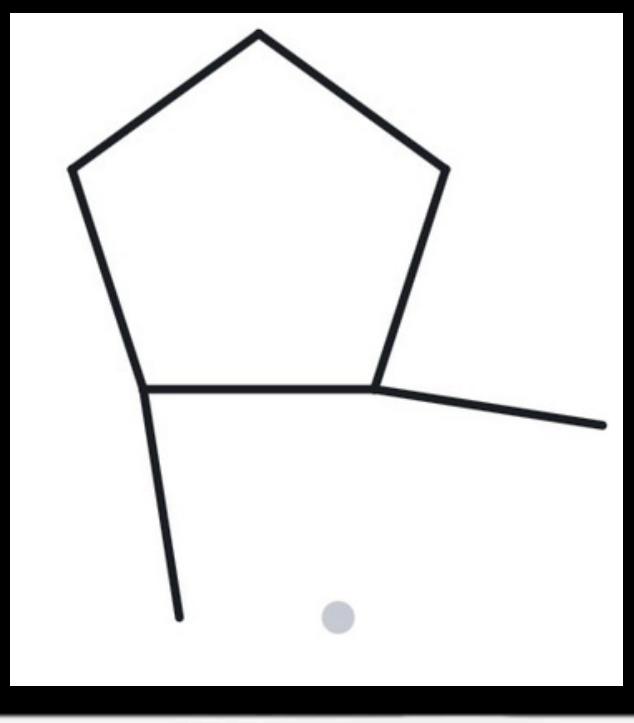
WORKED EXAMPLE B – NAME → STRUCTURE

B06 + DEVELOPING — GIVEN NAME: 1,2-dimethylcyclopentane

1. NAME DECODED

Parent: cyclopentane → a 5-carbon ring.
Substituents: two methyl groups on C1 and C2.
Expected formula: 5 + 2 = 7 C; one ring → $C_nH_{2n} \rightarrow C_7H_{14}$

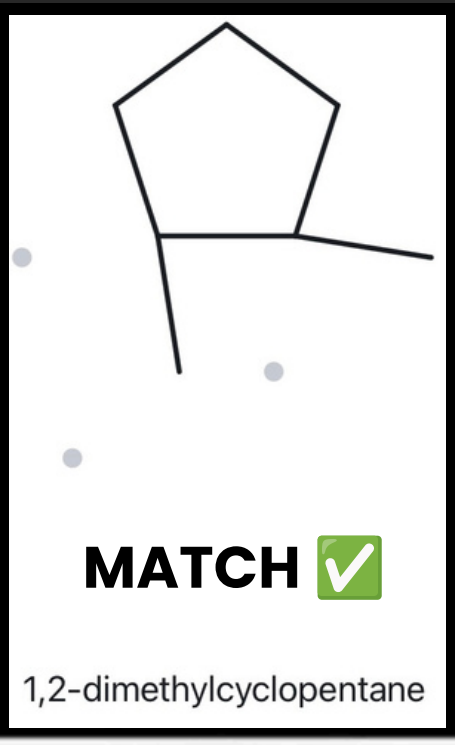
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: C_7H_{14}
Using the cycloalkane formula: C_nH_{2n}
$C_7H_2(_7) = C_7H_{14}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

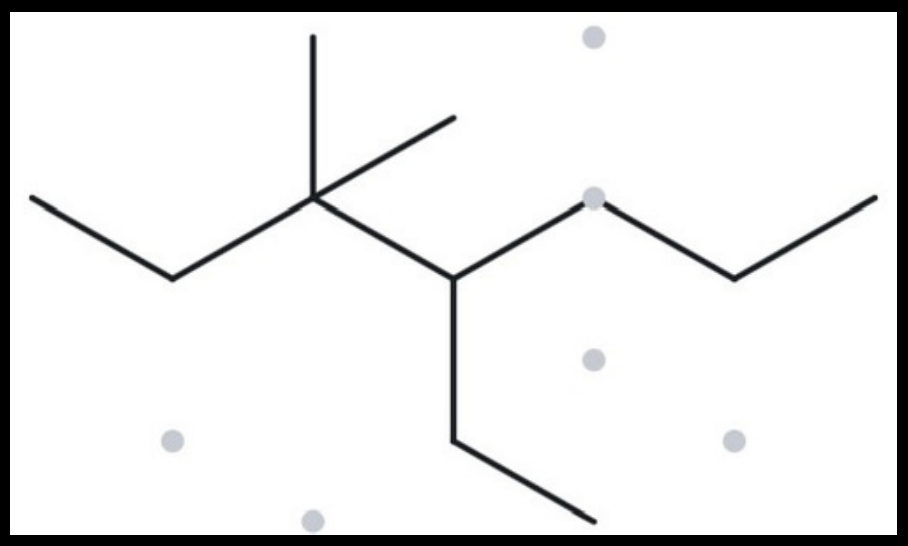
WORKED EXAMPLE B – NAME → STRUCTURE

B07 + DEVELOPING — GIVEN NAME: 4-ethyl-3,3-dimethylheptane

1. NAME DECODED

Parent: heptane → a 7-carbon chain.
Substituents: ethyl on C4; two methyl groups on C3.
Expected formula: 7 + 2 + 1 + 1 = 11 C; no ring → C_nH_{2n+2} → $C_{11}H_{24}$

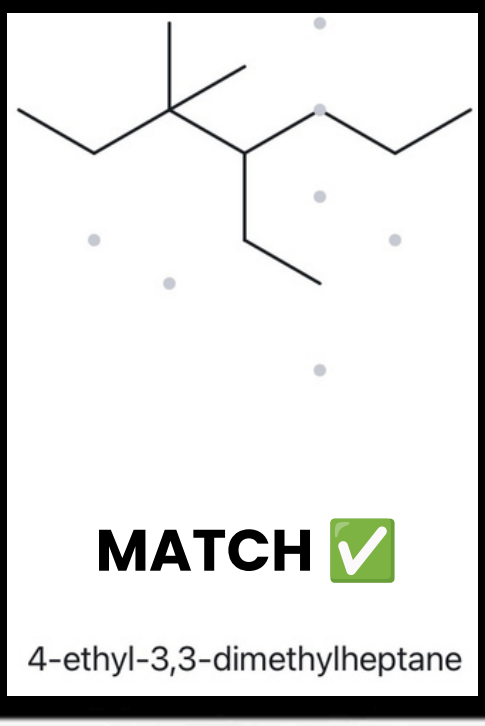
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$
Using the alkane formula: C_nH_{2n+2}
$C_{11}H_{2(11)+2} = C_{11}H_{24}$

4. BACK CHECK



VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

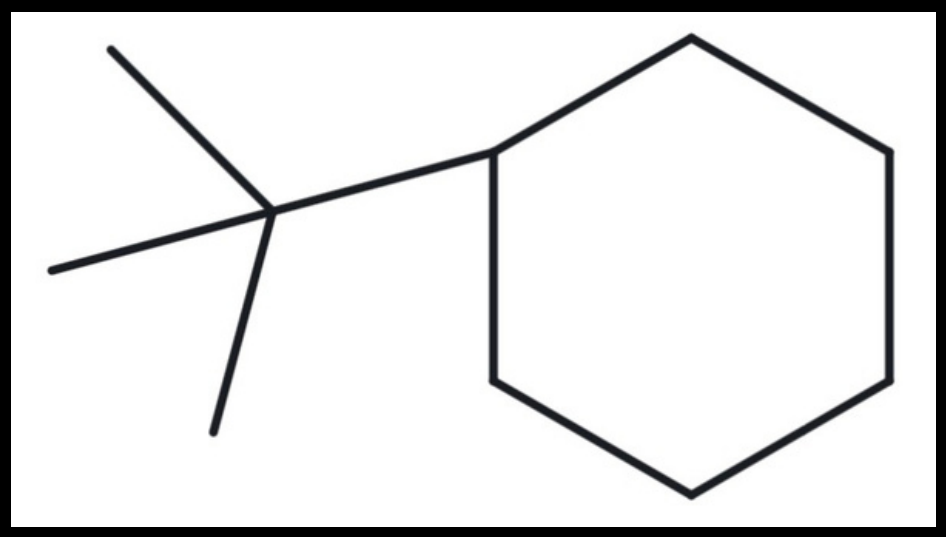
WORKED EXAMPLE B – NAME → STRUCTURE

B08 + CHALLENGE — GIVEN NAME: tert-butylcyclohexane

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: tert-butyl group attached to C1 of the ring.
Expected formula: 6 + 4 = 10 C; one ring → C_nH_{2n} → $C_{10}H_{20}$

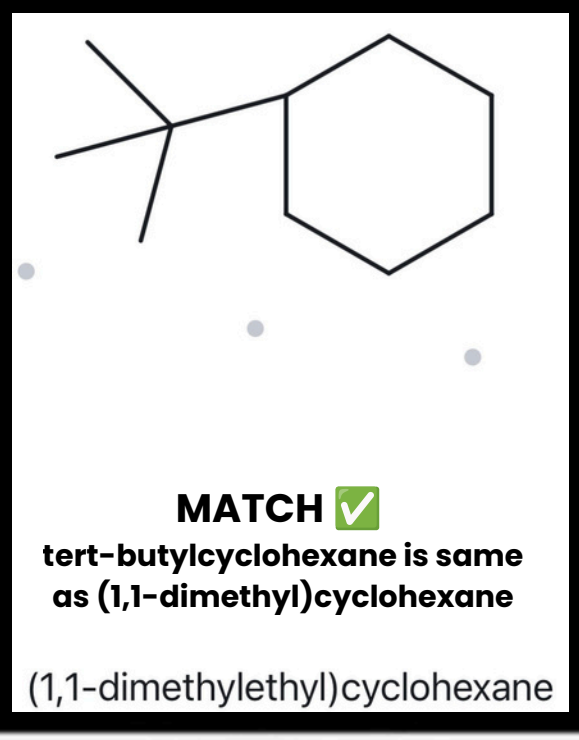
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
Using the cycloalkane formula: C_nH_{2n}
$C_{10}H_{2(10)} = C_{10}H_{20}$

4. BACK CHECK



MATCH ✅
tert-butylcyclohexane is same as (1,1-dimethyl)cyclohexane

(1,1-dimethylethyl)cyclohexane

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

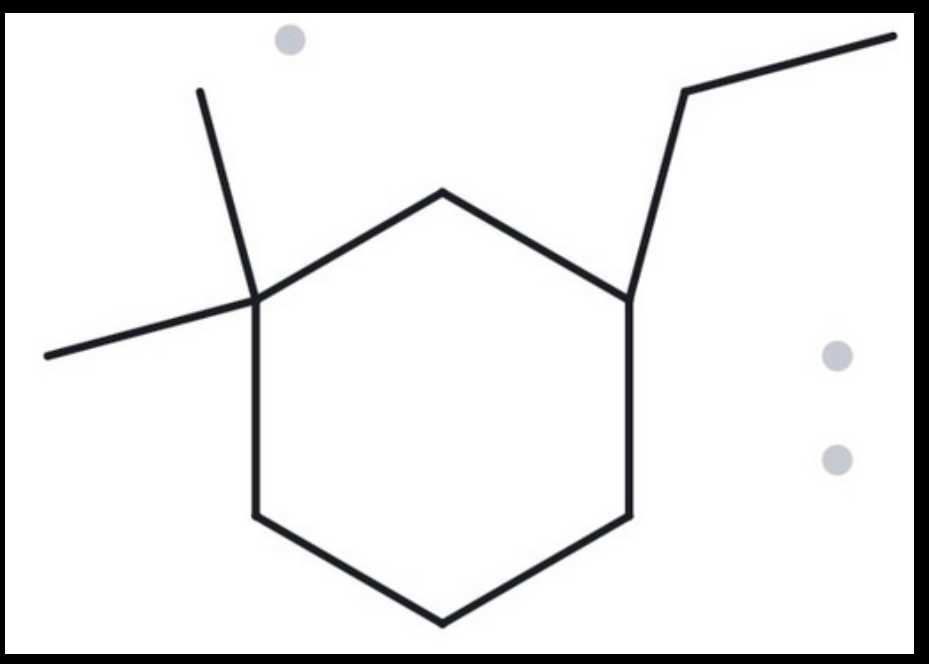
WORKED EXAMPLE B – NAME → STRUCTURE

B09 + CHALLENGE — GIVEN NAME: 3-ethyl-1,1-dimethylcyclohexane

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: ethyl on C3; two methyl groups on C1.
Expected formula: 6 + 2 + 1 + 1 = 10 C; one ring → C_nH_{2n} → $C_{10}H_{20}$

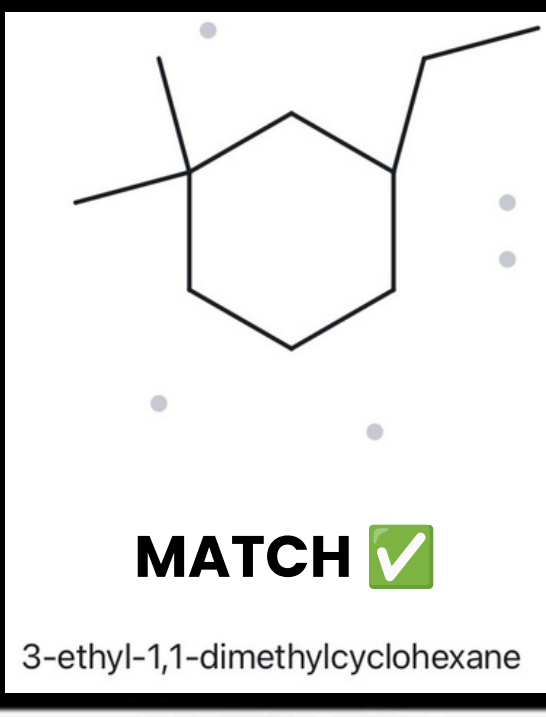
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
Using the cycloalkane formula: C_nH_{2n}
$C_{10}H_{2(10)} = C_{10}H_{20}$

4. BACK CHECK



MATCH ✅

3-ethyl-1,1-dimethylcyclohexane

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded

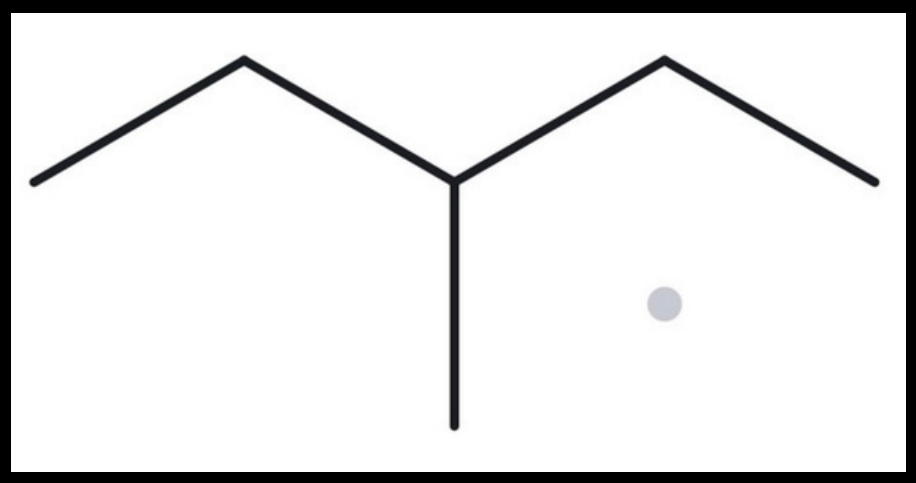
WORKED EXAMPLE B – NAME → STRUCTURE

BIO + CHALLENGE — GIVEN NAME: 2-ethylbutane

1. NAME DECODED

Given name: 2-ethylbutane → a 4-carbon parent chain (butane).
Longest chain: 5 consecutive carbon atoms → pentane.
Substituents: methyl on C3.
Correct IUPAC name: 3-methylpentane.
Expected formula: 5 + 1 = 6 C; no ring → C_nH_{2n+2} → C_6H_{14}

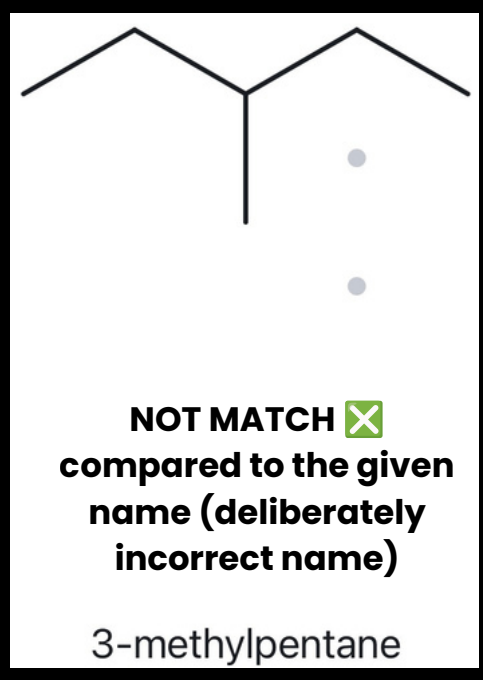
2. BOND LINE STRUCTURE



3. FORMULA CHECK

Total carbon atoms: 4 + 2 = 6 C.
Using the alkane formula: C_nH_{2n+2}
$C_6H_2(_6)+_2 = C_6H_{14}$

4. BACK CHECK



NOT MATCH ❎
compared to the given name (deliberately incorrect name)

3-methylpentane

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituesnt position match.
- ☑ formula agrees
- ☑ back-check recorded